

Sr. No.		Description	UOM (Wherever Applicable)	Data (Common For All Models)
A		General Points		
	1	Cooling Capacity	ton _R	Refer Teksel Software
	2	Power Consumption	kW	Refer Teksel Software
	3	Specific Power Consumption	kW/ton _R	Refer Teksel Software
	4	Co-Efficient of Performance (COP)	kW/kW	Refer Teksel Software
	5	No. of Compressors	Nos.	Contact To Engineering Team
	6	No. of Individual Refrigerant Circuits	Nos.	Contact To Engineering Team
	7	Refrigerant		
	i	Name	-	R134a
	ii	Quantity	kg	Refer ESP-18-19-007
	iii	Technical Specifications	-	Refer ESP-18-19-003
	8	Sound Pressure Level		
	i	Noise Level	dB	Refer ESP-15-16-104
	ii	Measuring Standard	-	ANSI/AHRI Standard 575-2008
	9	Insulation Details		
	i	Material	-	Closed Cell Nitrile Foam
	ii	Insulation Thickness on Various Parts	-	For Standard Temperature Range (LWT upto 3 °C)
		Evaporator Shell	mm	32
		Evaporator Tubesheet	mm	19
		Evaporator Dished End	mm	19
		Evaporator M.W.Box (If Applicable)	mm	19
		Evaporator Support Plate	mm	19
		Compressor Motor Body	mm	19
		Suction Line Assembly	mm	19
		Liquid Line Assembly	mm	9
	iv	Density	kg/m ³	76.6
	v	Thermal Conductivity	W/m.K	0.035 (at 0 °C Mean Temperature)
	vi	Standard	-	IS 14164
	vii	Adhesive	-	Blend of Synthetic Polymers and Synthetic Resin
	viii	Insulation Specifications	-	Refer ESP-18-19-004
	10	Vibration		
	i	Vibration Level	mm/sec	Less than 1.5 mm/sec
	ii	Vibration control	-	Rubber Pads (Standard) / Spring Isolators (At an Additional Cost)
	iii	Standard	-	IS 12075
	11	Painting Specification		
	i	Paint Type	-	RAL 7035
	ii	Standard	-	Coating as per KCPL Standards
	12	Overall Dimensions		
	i	Approx. Length	mm	Refer General Arrangement & Foundation Detail Drawing
	ii	Approx. Width	mm	Refer General Arrangement & Foundation Detail Drawing
	iii	Approx. Height	mm	Refer General Arrangement & Foundation Detail Drawing
	13	Space Clearances Required		
	i	Plain End Side (For Tube Cleaning)	mm	Refer General Arrangement & Foundation Detail Drawing
	ii	All Other Sides	mm	Refer General Arrangement & Foundation Detail Drawing
	iii	Overhead	mm	Refer General Arrangement & Foundation Detail Drawing
	14	Weight		
	i	Approx. Shipping Weight	kg	Refer General Arrangement & Foundation Detail Drawing
	ii	Approx. Operating Weight	kg	Refer General Arrangement & Foundation Detail Drawing
	15	Cable Sizes		
	i	Aluminum Cable	-	Refer ESP-14-15-01
	ii	Copper Cable	-	Refer ESP-14-15-01
B		Compressor Details		
	1	Make	-	Kirloskar Chillers Private Limited
	2	Type / Description	-	Semi-Hermetic Centrifugal Compressor
	3	Model	-	Refer Teksel Software
	4	Drive	-	Gear Driven
	5	Compressor Speed	RPM	Refer "KSC-R134a-02" Sheet
	6	Capacity Control Percentage	%	Refer Teksel Software
	7	Type of Capacity Control	-	Stepless
	8	Capacity Control Mechanism	-	IGV
	9	Design Pressure	bar	16
	10	Bearings		

	i	Types of Bearings	-	Hydrodynamic Bearings - For Radial Load and Thrust Load
	ii	Material of Construction	-	Aluminum
	iii	Class of Bearing	-	Proprietary Data
11		Lubrication		
	i	Type	-	Forced Lubrication by Oil Pump
	ii	Lubricating Oil	-	Synthetic Oil
	iii	Grade of Lubricating Oil	-	Proprietary Data
	iv	Quantity	Liter	Refer General Arrangement & Foundation Detail Drawing
12		Compressor Components MOC		
	i	Impeller	-	Aluminum
	ii	Casing	-	Cast Iron
	iii	Shaft	-	Alloy Steel
13		Physical Data of Compressor		
	i	Impeller Diameter	mm	Refer "KSC-R134a-02" Sheet
	ii	No. of Impeller Stages	Nos.	Single Stage
14		Oil Filter		
	i	Micron Rating	Micron	4
	ii	Material of Construction	-	Resin Impregnated Fibres Supported with Screen
	iii	Quantity	Nos.	1 No. per Compressor for KSC063-KSC087 2 Nos. per Compressor for KSC100-KSC126
15		Copressor Isolation Type		
	i	At Suction	-	No Isolation
	ii	At Discharge	-	Check Valve (NRV)
C		Compressor Motor Details		
1		Make	-	Kirloskar Approved Vendor
2		Motor Type	-	Semi-Hermetic Squirrel Cage Induction Motor
3		Type of Duty	-	Continuous
4		Motor Rating	kW	Consult with Engineering Department on Case to Case Basis
5		Motor Speed (Synchronous)	RPM	3000
6		Ingress Protection (IP)	-	NA, Being Semi-Hermetic Type
7		GD ² of Rotor	-	Proprietary Data
8		Whether SPDP or TEFC?	-	NA, Being Semi-Hermetic Type
10		Performance Indicators		
	i	Motor Efficiency Class	-	NA
	ii	Motor Power	kW	Consult with Engineering Department on Case to Case Basis
	iii	Motor Efficiency	-	Consult with Engineering Department on Case to Case Basis
	iv	Power Factor	-	Consult with Engineering Department on Case to Case Basis
	v	Class of Insulation	-	Class F
11		Motor Cooling		
	i	Motor Cooling Type	-	Refrigerant Cooled
	ii	Cooling Mechanism	-	By Liquid Refrigerant
	iii	Temperature at full load	°C	10 to 15 (At Normal Conditions)
12		Current Details		
	ii	Rated Load Current	A	Refer Teksel Software
	iii	Full Load Current	A	Consult with Engineering Department on Case to Case Basis
	iv	Inrush/Starting Current	A	Consult with Engineering Department on Case to Case Basis
	v	Locked Rotor Current	A	Consult with Engineering Department on Case to Case Basis
	vi	Starting Torque	N.m	Consult with Engineering Department on Case to Case Basis
	vii	No Load Current	A	Consult with Engineering Department on Case to Case Basis
13		Control Settings		
	i	No. of Starts per Hour	Nos.	4
	ii	Time Between STOP to START	Sec	300
	iii	Time Between START to START	Sec	900
E		Oil Sump and Pump Details		
1		Sump Make	-	Kirloskar Approved Vendor
2		Pump Details		
	i	Type	-	Submersible, G-Rotor Type
	ii	Make	-	Kirloskar Approved Vendor
	iii	Pump Capacity	LPM	45
	iv	Motor Type	-	Single Phase Induction Motor
	v	Motor Rating of Pump	HP	2.5
	vi	Motor Speed	RPM	3000
	vii	Power Supply	V/Hz	230V/50Hz/Single Phase
3		Material of Construction		
	i	Sump MOC	-	Cast Iron
4		Physical Details		

	i	Shell Diameter	inch	16
	ii	Approx. Height	mm	570
5		Oil Heater Details		
	i	Make	-	Kirloskar Approved Vendor
	ii	Quantity	Nos.	2 per Oil Sump
	iii	Power Supply	V	230
	iv	Rating	W	200
F		Oil Cooler		If Applicable
1		Type	-	Plate Type
2		Quantity	Nos.	One per Compressor
3		Heat Duty	kW	Depends on Working Conditions
4		Method of Cooling	-	Refrigerant Cooled
5		Material of Construction	-	Brazed PHE, Plate Material - SS
6		Pressure Drop		
	i	Oil Side	bar	less than 0.5
	ii	Refrigerant Side	bar	Proprietary Data
G		Evaporator Details		
1		Model	-	Refer Teksel Software
2		Design Code	-	As per KCPL Standards
3		Type	-	Shell and Tube Flooded Design
4		Tube Side (Fluid)	-	Chilled Water
5		Shell Side (Fluid)	-	Refrigerant
6		Design Parameters		
	i	Design Temperature (Refrigerant Side)	⁰ C	65
	ii	Max. Operating Pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
	iii	Design Pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
	iv	Test pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
	v	Testing method (Refrigerant Side)	-	Refer ESP-07-08-107
	vi	No. of Passes (Refrigerant Side)	Nos.	Single Pass
	vii	Design Temperature (Water Side)	⁰ C	65
	viii	Max. Operating Pressure (Water Side)	bar	Refer ESP-07-08-107
	ix	Design Pressure (Water Side)	bar	Refer ESP-07-08-107
	x	Test pressure (Water Side)	bar	Refer ESP-07-08-107
	xi	Testing method (Water Side)	-	Refer ESP-07-08-107
	xii	No. of Passes (Water Side)	Nos.	Two Pass
	xiii	Water Velocity	m/s	Less than 3 m/s
	xiv	Inlet Pressure (Water Side)	bar	Depends on Site Piping Layout (Maximum Allowable - 9.4 bar)
	xv	Evaporating Temperature	⁰ C	Consult with Engineering Department on Case to Case Basis
7		Physical Data of Evaporator		
	i	Overall Length of Evaporator	ft	Refer "HX Details" Sheet
	ii	Shell Diameter	inch	Refer "HX Details" Sheet
	iii	Shell Thickness	mm	Refer "HX Details" Sheet
	iv	Approx. Shell Length	mm	Refer "HX Details" Sheet
	v	Material of Construction of Shell	-	Mild Steel
	vi	Material Standard of Shell	-	Refer "MOC" Sheet
	vii	Tube Type/ Nature of Tube Surface	-	Integral Helical Fins on the Outside Surface and Integral Helical Ridges on the Inside Surface
	viii	Tube Length	mm	Refer "HX Details" Sheet
	ix	Tube Diameter	mm	Refer "HX Details" Sheet
	x	Tube Thickness	mm	Refer "HX Details" Sheet
	xi	Material of Construction of Tube	-	Cu
	xii	Material Standard of Tube	-	Refer "MOC" Sheet
	xiii	Water Volume in Evaporator	Liter	Refer Teksel Software
8		Water Box Details		
	i	Type	-	Standard - Dish Ends (M.W.Box - Optional)
	ii	Material	-	Mild Steel
	iii	Material Standard	-	Refer "MOC" Sheet
	iv	Nozzle size	NB	Refer Teksel Software
	v	End connection	-	Standard - Victaulic Conn. (Flanged Conn. - Optional)
	vi	MOC of Water Side Gasket	-	NAM AF 120
	vii	MOC of Refrigerant Side Gasket	-	NAM AF 159
9		Accessories Provided		
	i	Pressure Relief Valve	-	Spring Loaded (For Safety Valve Set Pressure Refer ESP)
	ii	Drain/Vent Valves	Inch	Plugged Connection Provided (3/8" NPT)
H		Condenser Details		
1		Model	-	Refer Teksel Software

	2	Design Code	-	As per KCPL Standards
	3	Type	-	Shell and Tube Flooded Design
	4	Tube Side (Fluid)	-	Chilled Water
	5	Shell Side (Fluid)	-	Refrigerant
	6	Design Parameters		
	i	Design Temperature (Refrigerant Side)	⁰ C	100
	ii	Max. Operating Pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
	iii	Design Pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
	iv	Test pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
	v	Testing method (Refrigerant Side)	-	Refer ESP-07-08-107
	vi	No. of Passes (Refrigerant Side)	Nos.	Single Pass
	vii	Design Temperature (Water Side)	⁰ C	100
	viii	Max. Operating Pressure (Water Side)	bar	Refer ESP-07-08-107
	ix	Design Pressure (Water Side)	bar	Refer ESP-07-08-107
	x	Test pressure (Water Side)	bar	Refer ESP-07-08-107
	xi	Testing method (Water Side)	-	Refer ESP-07-08-107
	xii	No. of Passes (Water Side)	Nos.	Two Pass
	xiii	Water Velocity	m/s	Less than 3 m/s
	xiv	Inlet Pressure	bar	Depends on Site Piping Layout (Maximum Allowable - 9.4 bar)
	xv	Condensing Temperature		Consult with Engineering Department on Case to Case Basis
	7	Physical Data of Condenser		
	i	Overall Length of Condenser	ft	Refer "HX Details" Sheet
	ii	Shell Diameter	inch	Refer "HX Details" Sheet
	iii	Shell Thickness	mm	Refer "HX Details" Sheet
	iv	Shell Length	mm	Refer "HX Details" Sheet
	v	Material of Construction of Shell	-	Mild Steel
	vi	Material Standard of Shell	-	Refer "MOC" Sheet
	vii	Tube Type/ Nature of Tube Surface	-	Integral Helical Fins on the Outside Surface and Integral Helical Ridges on the Inside Surface
	viii	Tube Length	mm	Refer "HX Details" Sheet
	ix	Tube Diameter	mm	Refer "HX Details" Sheet
	x	Tube Thickness	mm	Refer "HX Details" Sheet
	xi	Material of Construction of Tube	-	Cu
	xii	Material Standard of Tube	-	Refer "MOC" Sheet
	xiii	Water Volume in Condenser	Liter	Refer Teksel Software
	8	Water Box Details		
	i	Type	-	Standard - Dish Ends (M.W.Box - Optional)
	ii	Material	-	Mild Steel
	iii	Material Standard	-	Refer "MOC" Sheet
	iv	Nozzle size	NB	Refer Teksel Software
	v	End connection	-	Standard - Victaulic Conn. (Flanged Conn. - Optional)
	vi	MOC of Water Side Gasket	-	NAM AF 120
	vii	MOC of Refrigerant Side Gasket	-	NAM AF 159
	9	Accessories Provided		
	i	Pressure Relief Valve	-	Spring Loaded (For Safety Valve Set Pressure Refer ESP)
	ii	Drain/Vent Valves	Inch	Plugged Connection Provided (3/8" NPT)
I		Suction Line		
	1	Design Code	-	ASME B31.3
	2	Isolation Valve	-	No Isolation
	3	Material of Construction	-	Carbon Steel
	4	Material Standard	-	Refer "MOC" Sheet
	5	Angle Valve	-	Provided on Suction Line For Oil Recovery Line
J		Discharge Line		
	1	Design Code	-	ASME B31.3
	2	Isolation Valve	-	Check Valve (NRV)
	3	Material of Construction	-	Carbon Steel
	4	Material Standard	-	Refer "MOC" Sheet
	5	Skin Type Thermowell	-	Provided on Discharge Line For Discharge Temp. Sensor
K		Liquid Line		
	1	Design Code	-	ASME B31.3
	2	Expansion Valve		
	i	Type	-	Electronic Expansion Valve
	ii	Make	-	Refer "Make List" Sheet
	iii	Quantity	Nos.	Consult with Engineering Department on Case to Case Basis
	iv	Sight Glass	-	Inbuilt
	v	Moisture Indicator	-	NA

	3	Filter Drier	-	NA
	4	Material of Construction	-	Copper
	5	Material Standard	-	Refer "MOC" Sheet
N		Starter and Control Panel		
	1	Panel Enclosure	-	Starter and Control Panel Integrated in Single Fabricated Box
	2	Make	-	Kirloskar Approved Vendor
	3	Material of Enclosure	-	Rittal Enclosure - Sheet Steel Fabricated Enclosure - CRCA Sheet
	4	Thickness of Enclosure	mm	Rittal Enclosure - (Control Panel) Enclosure - 1.5 mm Door - 2 mm Fabricated Enclosure - (Starter Panel) Load Bearing Member - 2 mm Non-Load Bearing Member - 1.6 mm
	5	Ingress Protection (IP)	-	IP54
	6	Painting Specification		
	i	Paint Type	-	RAL 7035
	ii	Standard	-	Coating as per KCPL Standards
	7	Mounting Arrangement	-	Control Panel - Mounted on Chiller Starter Panel - Free Standing (Mounted on Chiller - Optional)
	8	Type of Starter	-	Star-Delta Starter (Soft Starter - Optional)
	9	Type of Isolation	-	MCCB for Star-Delta Starter FSD for Soft Starter
	10	Type of Protection	-	MCCB for Star-Delta Starter FSD for Soft Starter
	11	Switchgear Make	-	Siemens
	12	Electrical and Control Cables	-	Power - PVC Insulated Single Core (Vtg. Grade 1.1 kV) Control- PVC Insulated Single Core, Multicore Cable (Vtg. Grade 1.1 kV) Signal- Shielded Cable
	13	Optional Features		
	i	Phase Indicating Lamps	-	Special-Optional
	ii	Hooter	-	Special-Optional
	iii	Energymeter	-	Special-Optional
	iv	Door Handle	-	Special-Optional
	v	LOTO Arrangement	-	Special-Optional
O		Controller		
	1	Make	-	Refer "Make List" Sheet
	2	Transmitters	-	NA
	3	Oil Level Switch	-	Yes, Provided
	4	Oil Level Failure Trip	-	Yes, Provided
	5	LP Switch and Gauge	-	No, Controller Program will Take Care of Low Pressure
	6	HP Switch and Gauge	-	No, Controller Program will Take Care of High Pressure
	7	Chilled Water Flow Failure	-	Yes
	8	Cooling Water Flow Failure	-	Yes
	9	Reverse Rotor Protection	-	No
	10	High/Low Voltage Trip	-	Yes
	11	Low Current Trip (Current Based-Analog)	-	Yes
	12	High Current Trip (Current Based-Analog)	-	Yes
	13	Phase Failure/Reverse Phasing Trip	-	Yes
	14	Earth Fault Trip	-	No
	15	Communication Through RS232/RS485	-	RS485
	16	Display of Microprocessor	-	Yes
	17	Type of Display	-	13" Touch Screen Display
	18	Remote Monitoring Facility	-	Yes
	19	Output to DCS	-	Applicable (Only if RS485 is Available)